

Manvendra Singh Solanki

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EDUCATION

SRM University

Aug 2022 – May 2026

B.Tech in Computer Engineering (AI-ML), CGPA: 8.31

TECHNICAL SKILLS

Languages: Python, Java, HTML5, CSS3, JavaScript, C++

Backend & Tools: RESTful APIs, GitHub, Git, MySQL, MongoDB, AWS, SQL

Libraries: FastAPI, Pandas, NumPy, PyTorch, OpenCV, MediaPipe

Core Concepts: Object-Oriented Programming (OOP), Software Development Life Cycle (SDLC), Data Structures & Algorithms (DSA), Agile Methodologies, TCP/IP, IP Tables, Routing, ARP

PROFESSIONAL EXPERIENCE

CorporateServe Solution Pvt. Ltd.

Jun 2024 – Jul 2024

Web Developer | Intern | Gurgaon

- Collaborated directly with 10+ client stakeholders to translate technical requirements into engineering solutions, coordinating across client and infrastructure teams to ensure timely delivery.
- Developed and maintained REST APIs with MySQL, achieving system integration across 3 web applications with 30% reliability improvement and zero unplanned downtime across production infrastructure.
- Transformed ambiguous technical requirements from 10+ clients into effective front-end solutions, leading to a 30% boost in delivery efficiency and increased user interface performance.

PROJECTS

Nexus | Python, C++, HNSW, KD-Tree, RAG Pipeline, Ollama, FastAPI, PCA | [GitHub](#)

Apr 2026

- Built an AI-powered vector database in Python with HNSW indexing achieving $O(\log N)$ search over 768-dimensional LLM embeddings, replicating core logic of Pinecone & Weaviate.
- Deployed a local RAG pipeline (nomic-embed-text + llama3.2) auto-chunking docs into 250-word segments, retrieving top-3 semantic matches with zero API cost.
- Compared 3 search algorithms $\times$ 3 distance metrics via a live AI benchmarking dashboard with 2D PCA visualization, reducing algorithm selection time by 3x over manual testing.

Deep Packet Inspector | Python, Java, Multithreading, DPI, PyTest, Claude | [GitHub](#)

Mar 2026

- Engineered a Python-based multi-threaded DPI engine across a 4-stage pipeline using thread-safe queues and $O(1)$ 5-tuple hashing for scalable, flow-aware packet processing.
- Implemented TCP/IP stack parsing across 4 network layers (Ethernet, IPv4, TCP/UDP, TLS) with SNI extraction to classify HTTPS traffic into 5+ app categories, with schema validation to prevent errors.
- Designed a stateful flow engine with 3 rule types (IP, domain, app) for 100% session-level blocking; wrote per-stage unit tests and applied CI-style validation to catch regressions.

SERENE | Python, MediaPipe, OpenCV, Random-Forest algorithm, TensorFlow | [GitHub](#)

Oct 2025

- Built an end-to-end AI sign language recognition system in Python, integrating MediaPipe for real-time hand-landmark extraction and a trained Random Forest classifier for gesture recognition, delivering sub-second translation for 466M+ deaf or hard-of-hearing users globally.
- Expanded the gesture library by 15% and validated model accuracy through iterative testing cycles, refining the classification pipeline based on failure analysis and edge-case evaluation.
- Optimized end-to-end inference latency to under 200ms with auto-reset after every 6 words, ensuring real-time usability and accessibility at scale.

ACHIVEMENTS AND CERTIFICATIONS

- **Data Structures & Algorithms:** Solved 210+ problems on LeetCode, holds a 600+ LeetCode contest rating, and currently has a 2-Star rating on CodeChef.
- **Google AI Essentials & Kaggle Machine Learning Certification:** Completed foundational AI, prompt engineering, and hands-on machine learning training, applying data analysis techniques in practical projects.